

Conditional Things in ML for Mathematicians

A Rigorous Handbook of the Conditional Notation
Used in Machine-Learning Theses

A measure-theoretic reading of
 $\mathbb{P}(A|B), \quad \mathbb{E}[X|\mathcal{G}], \quad p(x|y), \quad X \perp\!\!\!\perp Y|Z, \quad H(X|Y), \quad \nabla_x \log p_t(x|y).$

From conditional probability to classifier-free guidance,
with every bar “|” given a definition.

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Preface

Machine-learning papers are written in a dialect of probability theory in which the vertical bar “|” is the busiest symbol on the page. A single thesis may, within three paragraphs, write the conditional density $p(x|y)$, the conditional expectation $\mathbb{E}[X|Y]$, the amortized variational posterior $q_\phi(z|x)$, the policy $\pi(a|s)$, and the conditional score $\nabla_x \log p_t(x|y)$ — and treat all of them as if their meaning were self-evident. For a reader trained in measure theory this is disorienting: the same glyph denotes a number, a function, a random variable, a Markov kernel, and a Radon–Nikodym derivative, depending on context that is rarely stated.

This handbook is a translation manual. Its purpose is to take every common piece of *conditional* notation that a mathematician will meet in an ML thesis and attach to it a single, unambiguous, measure-theoretic definition, together with the theorem that guarantees the object exists, the sense in which it is unique, and the pitfalls that the informal notation hides.

Who this is for. The intended reader knows graduate real analysis and measure-theoretic probability at the level of, say, the first chapters of Williams’ *Probability with Martingales* or Durrett’s *Probability: Theory and Examples*: σ -algebras, the Lebesgue integral, product measures, Radon–Nikodym, and the basic convergence theorems are assumed. No prior machine learning is assumed. Where an ML construction (a VAE, a diffusion model, an MDP) is needed, it is introduced from its probabilistic skeleton only, with enough detail to fix the notation and not one symbol more.

What this is not. It is not a machine-learning textbook, an optimization reference, or a survey of architectures. Approximation, estimation, and computation — the parts of ML that are genuinely about *learning* — appear only insofar as they are needed to explain what a symbol denotes. We are concerned with what the notation *means*, not with how well a network fits it.

Standing conventions. Unless stated otherwise, $(\Omega, \mathcal{F}, \mathbb{P})$ is a fixed probability space; random variables are measurable maps out of it; “a.s.” means $\mathbb{P}$ -almost surely; and all named random variables are assumed to take values in standard Borel (Polish, with their Borel σ -algebra) spaces, which is exactly the hypothesis that makes regular conditional distributions exist (Chapter 4). Densities are taken with respect to a reference measure — Lebesgue on $\mathbb{R}^d$ or counting measure on a discrete set — that we name when it matters and suppress when it does not. A short index of recurring symbols is collected in Appendix B.

How to read it. Parts I–V build the rigorous objects in order of dependence: conditional expectation, then conditional distributions and densities, then conditional independence, then the information-theoretic conditionals. Part VI is a dictionary: each entry takes a symbol *as it is actually printed in papers* and points back to the definition that licenses it. A reader who only wants to decode a particular formula can start there and follow the cross-references inward.

A note on rigor and honesty. Several identities that ML papers write with an equals sign are, strictly, only true almost surely, or only after a regular version has been fixed, or only under

an absolute-continuity hypothesis that is left implicit. Rather than smooth these over, the handbook flags them, because the gap between “=” and “ $\stackrel{\text{a.s.}}{=}$ ” is exactly where a mathematician’s intuition is most likely to be misled.

Part I

Foundations and Conditional Expectation

Chapter 1

Why elementary conditioning is not enough

1.1 The schoolbook definition and where it breaks

Every reader knows the elementary definition.

Definition 1.1 (Elementary conditional probability). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $B \in \mathcal{F}$ an event with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$,

$$\mathbb{P}(A | B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (1.1)$$

For fixed B with $\mathbb{P}(B) > 0$ the set function $A \mapsto \mathbb{P}(A | B)$ is a bona fide probability measure on $(\Omega, \mathcal{F})$, and $\mathbb{E}[X | B] := \mathbb{P}(B)^{-1} \int_B X \, d\mathbb{P}$ is its expectation operator. Nothing here is subtle. The trouble is that essentially every conditional quantity in machine learning conditions on an event of probability zero.

Example 1.2 (The event we actually want has probability zero). Let (X, Y) be jointly continuous on $\mathbb{R}^2$, e.g. $Y \sim \mathcal{N}(0, 1)$ and $X | Y$ Gaussian. The object an ML paper denotes $p(x | y)$, $\mathbb{E}[X | Y = y]$, or $\mathbb{P}(X \in A | Y = y)$ conditions on the event $\{Y = y\}$, which has $\mathbb{P}(Y = y) = 0$. Formula (1.1) returns $\frac{0}{0}$. The symbol is in universal use and the schoolbook definition cannot assign it a value.

The naive repair — “condition on $\{y - \varepsilon < Y < y + \varepsilon\}$ and let $\varepsilon \rightarrow 0$ ” — can be made to work, but its answer depends on *how* the limiting family of events is chosen. That dependence is not a technicality; it is a genuine ambiguity with a name.

Pitfall (The Borel–Kolmogorov paradox, previewed). Conditioning on a measure-zero event $\{Y = y\}$ has no answer until one specifies the *family* of shrinking events used to approach it. Different parameterizations of the same null event yield different “conditional distributions.” We treat this carefully in Section 5.3; for now it is the warning that motivates everything below. The resolution is to stop conditioning on a single event and instead condition on a σ -algebra, obtaining an object defined only up to almost-sure equivalence.

1.2 The shift in viewpoint

Measure-theoretic conditioning makes three moves at once, and a mathematician new to the ML literature should hold all three in mind.

- (i) **Condition on information, not on an event.** The primitive is a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ — “everything we are allowed to know.” Conditioning on a random variable Y means conditioning on $\mathcal{G} = \sigma(Y)$.

- (ii) **The answer is a random variable, not a number.** $\mathbb{E}[X | \mathcal{G}]$ is itself a $\mathcal{G}$ -measurable random variable; evaluating it “at $Y = y$ ” is a secondary, and more delicate, operation (Section 3.1).
- (iii) **Identities hold almost surely.** The defining property pins down $\mathbb{E}[X | \mathcal{G}]$ only up to a $\mathbb{P}$ -null set. Every “=” between conditional objects is, by default, an “ $\stackrel{\text{a.s.}}{=}$ ”.

The remainder of Part I develops move (i)–(iii) into Kolmogorov’s definition of conditional expectation, from which conditional probability, conditional variance, and (in Part III) the conditional densities and kernels of the ML literature are all derived as special cases.

Part II

Conditional Expectation

Chapter 2

Conditional expectation given a σ -algebra

2.1 Kolmogorov's definition

Definition 2.1 (Conditional expectation). Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. A *version of the conditional expectation of X given $\mathcal{G}$* is any random variable Z satisfying

(CE1) Z is $\mathcal{G}$ -measurable;

(CE2) $Z \in L^1(\mathbb{P})$, and for every $G \in \mathcal{G}$,

$$\int_G Z \, d\mathbb{P} = \int_G X \, d\mathbb{P}. \quad (2.1)$$

Any such Z is written $\mathbb{E}[X | \mathcal{G}]$.

Property (CE1) says Z uses only the information in $\mathcal{G}$; property (2.1) says Z has the same “local averages” as X over every event we can resolve with $\mathcal{G}$. Note what is *not* required: no pointwise relation between $Z(\omega)$ and $X(\omega)$ holds, only an averaged one.

Theorem 2.2 (Existence and a.s. uniqueness). *For every $X \in L^1(\mathbb{P})$ and every sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ a version $\mathbb{E}[X | \mathcal{G}]$ exists, and any two versions agree $\mathbb{P}$ -almost surely.*

Proof sketch. Existence. Assume first $X \geq 0$. Define a finite measure on $(\Omega, \mathcal{G})$ by $\nu(G) := \int_G X \, d\mathbb{P}$. Then $\nu \ll \mathbb{P}|_{\mathcal{G}}$, since $\mathbb{P}(G) = 0 \Rightarrow \nu(G) = 0$. By the Radon–Nikodym theorem (Theorem A.1) applied on the measurable space $(\Omega, \mathcal{G})$ there is a $\mathcal{G}$ -measurable density $Z = d\nu/d(\mathbb{P}|_{\mathcal{G}})$, which is exactly (2.1). For general X set $\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X^+ | \mathcal{G}] - \mathbb{E}[X^- | \mathcal{G}]$.

Uniqueness. If Z, Z' both satisfy (2.1), then $\int_G (Z - Z') \, d\mathbb{P} = 0$ for all $G \in \mathcal{G}$. Taking $G = \{Z - Z' > 0\} \in \mathcal{G}$ forces $\mathbb{P}(Z > Z') = 0$, and symmetrically; hence $Z \stackrel{\text{a.s.}}{=} Z'$. $\square$

Convention 2.3 (“The” conditional expectation). Because versions differ only on a null set, one speaks of *the* conditional expectation and writes $\mathbb{E}[X | \mathcal{G}]$ for an arbitrary version, with all ensuing equalities understood $\mathbb{P}$ -a.s. When a statement is made for a *specific* version (as in the regular conditional distributions of Chapter 4), we say so explicitly.

Remark 2.4 (L^2 projection picture). If $X \in L^2(\mathbb{P})$ then $L^2(\Omega, \mathcal{G}, \mathbb{P})$ is a closed subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$, and $\mathbb{E}[X | \mathcal{G}]$ is exactly the orthogonal projection of X onto it. Equivalently it solves

$$\mathbb{E}[X | \mathcal{G}] = \arg \min_{\substack{Z \text{ } \mathcal{G}\text{-meas.} \\ Z \in L^2}} \mathbb{E}[(X - Z)^2], \quad (2.2)$$

the *best mean-square predictor of X from the information $\mathcal{G}$* . This is the identity that ML’s “the regression function $\mathbb{E}[Y | X = \cdot]$ minimizes squared error” is quoting, and it is the cleanest one-line answer to “what is conditional expectation?” for an analyst: it is a projection.

2.2 The calculus of conditional expectation

The following properties are used constantly — usually silently — in ML derivations (the ELBO, policy-gradient identities, the tower rule behind bootstrapping in reinforcement learning). All equalities are $\mathbb{P}$ -a.s.; all random variables are in L^1 unless a moment is named.

Proposition 2.5 (Elementary properties). *Let $\mathcal{G}, \mathcal{H} \subseteq \mathcal{F}$ be sub- σ -algebras and $a, b \in \mathbb{R}$.*

- (1) (Linearity) $\mathbb{E}[aX + bY | \mathcal{G}] = a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}]$.
- (2) (Monotonicity) $X \leq Y$ a.s. $\Rightarrow \mathbb{E}[X | \mathcal{G}] \leq \mathbb{E}[Y | \mathcal{G}]$; in particular $|\mathbb{E}[X | \mathcal{G}]| \leq \mathbb{E}[|X| | \mathcal{G}]$.
- (3) (Total expectation) $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X]$, taking $\mathcal{G} = \Omega$ in (2.1).
- (4) (Tower / iterated expectations) if $\mathcal{H} \subseteq \mathcal{G}$ then

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}] = \mathbb{E}[\mathbb{E}[X | \mathcal{H}] | \mathcal{G}]. \quad (2.3)$$

- (5) (Taking out what is known) if Y is $\mathcal{G}$ -measurable and $XY \in L^1$ (e.g. Y bounded), then

$$\mathbb{E}[YX | \mathcal{G}] = Y \mathbb{E}[X | \mathcal{G}]. \quad (2.4)$$

- (6) (Independence drops the conditioning) if $\sigma(X)$ is independent of $\mathcal{G}$, then $\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X]$ (a constant).

- (7) (Conditional Jensen) if $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $\varphi(X) \in L^1$, then

$$\varphi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\varphi(X) | \mathcal{G}]. \quad (2.5)$$

With $\varphi(t) = |t|^p$ ($p \geq 1$) this gives the conditional L^p contraction $\|\mathbb{E}[X | \mathcal{G}]\|_p \leq \|X\|_p$.

Notes on proof. (1)–(3) are immediate from Definition 2.1. For the tower rule, the left equality holds because $\mathbb{E}[X | \mathcal{H}]$ is $\mathcal{H}$ -measurable and satisfies (2.1) over $\mathcal{H} \subseteq \mathcal{G}$; the right equality is (2.1) read for $\mathcal{H}$. Property (5) is proved first for $Y = \mathbf{1}_{G_0}$, $G_0 \in \mathcal{G}$ (both sides integrate to $\int_{G \cap G_0} X$ over $G \in \mathcal{G}$), then extended by linearity and monotone/dominated convergence. Property (7) follows from the unconditional Jensen inequality applied to the probability measures $\mathbb{P}(\cdot | \mathcal{G})$ of Chapter 4, or directly via suprema of affine minorants of φ . $\square$

Remark 2.6 (Why ML cares about the tower rule). The identity (2.3) is the backbone of *bootstrapping*. In reinforcement learning the value function satisfies $V(s) = \mathbb{E}[r + \gamma V(S') | S = s]$ precisely because a total-reward conditional expectation can be peeled one step at a time by the tower rule; see Section 9.3. In variational inference the same rule lets one push an expectation through a latent variable to derive the ELBO (Section 8.3).

2.3 Conditional probability and conditional variance

Definition 2.7 (Conditional probability given $\mathcal{G}$). For $A \in \mathcal{F}$ define the random variable

$$\mathbb{P}(A | \mathcal{G}) := \mathbb{E}[\mathbf{1}_A | \mathcal{G}]. \quad (2.6)$$

By Proposition 2.5, for each *fixed* A we have $0 \leq \mathbb{P}(A | \mathcal{G}) \leq 1$ a.s., $\mathbb{P}(\Omega | \mathcal{G}) = 1$ a.s., and countable additivity holds a.s. for any fixed countable disjoint family. The crucial subtlety — that these “a.s.” qualifiers cannot in general be made simultaneous over *all* A — is the entire content of Chapter 4.

Definition 2.8 (Conditional variance). For $X \in L^2(\mathbb{P})$,

$$\text{Var}(X | \mathcal{G}) := \mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])^2 | \mathcal{G}] = \mathbb{E}[X^2 | \mathcal{G}] - (\mathbb{E}[X | \mathcal{G}])^2. \quad (2.7)$$

Proposition 2.9 (Law of total variance). For $X \in L^2(\mathbb{P})$,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | \mathcal{G})] + \text{Var}(\mathbb{E}[X | \mathcal{G}]). \quad (2.8)$$

Proof. Expand both terms using the tower rule and $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X]$; the cross term vanishes because $\mathbb{E}[X | \mathcal{G}]$ is the L^2 -projection of X (Remark 2.4), so $X - \mathbb{E}[X | \mathcal{G}] \perp L^2(\mathcal{G}) \ni \mathbb{E}[X | \mathcal{G}] - \mathbb{E}[X]$. $\square$

Remark 2.10. Equation (2.8) is the “bias/variance” bookkeeping behind, e.g., the decomposition of predictive uncertainty into *aleatoric* ($\mathbb{E}[\text{Var}(X | \mathcal{G})]$, irreducible noise) and *epistemic* ($\text{Var}(\mathbb{E}[X | \mathcal{G}])$, reducible by more information) parts that recurs throughout the Bayesian deep-learning literature.

Chapter 3

Conditioning on a random variable

3.1 From $\sigma(Y)$ to a function of y

ML papers almost never write $\mathbb{E}[X | \mathcal{G}]$; they write $\mathbb{E}[X | Y]$ and, more often, $\mathbb{E}[X | Y = y]$. Making sense of the second requires one structural theorem.

Definition 3.1 (Conditioning on a random variable). For a random variable $Y : \Omega \rightarrow \mathcal{Y}$ set $\sigma(Y) := Y^{-1}(\mathcal{B}_{\mathcal{Y}})$ and define $\mathbb{E}[X | Y] := \mathbb{E}[X | \sigma(Y)]$, $\mathbb{P}(A | Y) := \mathbb{P}(A | \sigma(Y))$.

Lemma 3.2 (Doob–Dynkin). *Let $Y : \Omega \rightarrow \mathcal{Y}$ and $Z : \Omega \rightarrow \mathbb{R}$ be measurable, with $\mathcal{Y}$ standard Borel. Then Z is $\sigma(Y)$ -measurable if and only if there exists a Borel function $g : \mathcal{Y} \rightarrow \mathbb{R}$ with $Z = g(Y)$.*

Applied to $Z = \mathbb{E}[X | Y]$, which is $\sigma(Y)$ -measurable by construction, the lemma produces a Borel g with $\mathbb{E}[X | Y] = g(Y)$ a.s. *This g is what licenses the notation $\mathbb{E}[X | Y = y]$:*

Definition 3.3 (The evaluated conditional expectation). With g as above, define

$$\mathbb{E}[X | Y = y] := g(y), \quad y \in \mathcal{Y}. \quad (3.1)$$

Warning 3.4 (What “ $\mathbb{E}[X | Y = y]$ ” really is). The function g of Lemma 3.2 is determined only up to a P_Y -null set, where $P_Y = \mathbb{P} \circ Y^{-1}$ is the law of Y . Hence $\mathbb{E}[X | Y = y]$ is well defined only for P_Y -almost every y , and its value at any single, pre-specified y_0 is meaningless in isolation. Two consequences a mathematician should hold onto:

- Statements like “ $\mathbb{E}[X | Y = y] = h(y)$ for all y ” are abuses: they assert equality of two versions, true only P_Y -a.e.
- The random variable $\mathbb{E}[X | Y] = g(Y)$ is well defined a.s., because Y never visits the P_Y -null exceptional set. The pathology lives entirely in the act of plugging in a deterministic y .

Theorem 3.5 (Change of variables / the law of $\mathbb{E}[X | Y]$). *With $g(y) = \mathbb{E}[X | Y = y]$, for every bounded Borel $\psi : \mathcal{Y} \rightarrow \mathbb{R}$,*

$$\mathbb{E}[X \psi(Y)] = \mathbb{E}[g(Y) \psi(Y)] = \int_{\mathcal{Y}} g(y) \psi(y) P_Y(dy). \quad (3.2)$$

Proof. The first equality is (2.4) with the $\sigma(Y)$ -measurable factor $\psi(Y)$, then total expectation; the second is the image-measure (“law of the unconscious statistician”) formula. $\square$

Equation (3.2) is the rigorous skeleton of the ubiquitous ML move $\mathbb{E}_{X,Y}[\cdot] = \mathbb{E}_Y[\mathbb{E}_{X|Y}[\cdot]]$, written in papers as “condition on Y , then average.”

3.2 The regression function

Definition 3.6 (Regression function). For $X \in L^1$ the map $m(y) := \mathbb{E}[X | Y = y]$ is the *regression function* of X on Y .

By Remark 2.4, when $X \in L^2$ the random variable $m(Y)$ is the $L^2(\sigma(Y))$ projection of X , hence

$$m = \arg \min_{f \text{ Borel}, f(Y) \in L^2} \mathbb{E}[(X - f(Y))^2]. \quad (3.3)$$

This is precisely the population object that a regression network with the squared-error loss is trying to approximate; the Bayes-optimal predictor of X from Y under squared loss *is* the conditional expectation. Different losses target different conditional functionals: absolute loss targets a conditional median, and the “pinball” loss at level τ targets the conditional τ -quantile — a point worth keeping in mind when a paper says a network “predicts Y given X ” without naming the loss.

Part III

Conditional Distributions, Kernels, and Densities

Chapter 4

Markov kernels and regular conditional distributions

Conditional expectation gave us, for each *fixed* event A , a random variable $\mathbb{P}(A | \mathcal{G})$. What an ML paper means by “the conditional distribution $p(\cdot | y)$ ” is the far stronger object in which A and ω vary together and the result is, for almost every outcome, an honest probability measure. The bridge is the notion of a Markov kernel.

4.1 Markov kernels

Definition 4.1 (Markov kernel). Let $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$ and $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ be measurable spaces. A *Markov kernel* (or *transition kernel*, *stochastic kernel*, *conditional distribution*) from $\mathcal{Y}$ to $\mathcal{X}$ is a map $\kappa : \mathcal{Y} \times \mathcal{B}_{\mathcal{X}} \rightarrow [0, 1]$ such that

(K1) for each fixed $y \in \mathcal{Y}$, $A \mapsto \kappa(y, A)$ is a probability measure on $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$;

(K2) for each fixed $A \in \mathcal{B}_{\mathcal{X}}$, $y \mapsto \kappa(y, A)$ is $\mathcal{B}_{\mathcal{Y}}$ -measurable.

We write $\kappa(y, dx)$ or $\kappa(dx | y)$, and overload notation by writing $\kappa(y, \cdot) =: p(\cdot | y)$ once a reference measure is fixed.

A Markov kernel is exactly “a measurably-indexed family of probability measures.” It is the rigorous content of every conditional distribution symbol in the ML literature: $p_{\theta}(x | y)$, $q_{\phi}(z | x)$, $\pi(\cdot | s)$, the forward and reverse transitions of a diffusion, and the rows of a Markov-chain transition matrix are all Markov kernels.

Remark 4.2 (Kernels compose; this is why “stacking conditionals” works). Given a measure μ on $\mathcal{Y}$ and a kernel κ from $\mathcal{Y}$ to $\mathcal{X}$, one builds a measure on the product and a marginal:

$$(\mu \otimes \kappa)(B \times A) = \int_B \kappa(y, A) \mu(dy), \quad (\mu \kappa)(A) = \int_{\mathcal{Y}} \kappa(y, A) \mu(dy). \quad (4.1)$$

Kernels also compose: $(\kappa_2 \kappa_1)(y, A) = \int \kappa_2(x, A) \kappa_1(y, dx)$ is the Chapman–Kolmogorov rule. This algebra is what makes an autoregressive factorization $p(x_{1:T}) = \prod_t p(x_t | x_{<t})$ or a Markov chain $p(x_{0:T}) = p_0 \prod_t \kappa(x_{t-1}, dx_t)$ a genuine probability measure on the product space rather than a formal product of symbols (Sections 9.1, 9.2).

4.2 Regular conditional distributions

Definition 4.3 (Regular conditional distribution). Let $X : \Omega \rightarrow \mathcal{X}$ and $Y : \Omega \rightarrow \mathcal{Y}$ be random variables. A *regular conditional distribution* (r.c.d.) of X given Y is a Markov kernel κ from $\mathcal{Y}$

to $\mathcal{X}$ such that for every $A \in \mathcal{B}_{\mathcal{X}}$,

$$\kappa(Y, A) \stackrel{\text{a.s.}}{=} \mathbb{P}(X \in A | Y). \quad (4.2)$$

Equivalently, for all $A \in \mathcal{B}_{\mathcal{X}}$, $B \in \mathcal{B}_{\mathcal{Y}}$,

$$\mathbb{P}(X \in A, Y \in B) = \int_B \kappa(y, A) P_Y(dy). \quad (4.3)$$

We then write $P_{X|Y}(A|y) := \kappa(y, A)$.

The point of (4.2) is the order of quantifiers. For each fixed A the right-hand side is only an a.s.-defined random variable; the r.c.d. asserts the existence of a *single* kernel that simultaneously serves as a version for *all* A at once, *and* is a genuine probability measure in A for every y . That such a reconciliation is possible is not automatic.

Theorem 4.4 (Existence on standard Borel spaces). *If $\mathcal{X}$ is a standard Borel space (a Borel subset of a Polish space with its Borel σ -algebra), then a regular conditional distribution of X given Y exists, and it is unique in the sense that any two r.c.d.s κ, κ' satisfy $\kappa(y, \cdot) = \kappa'(y, \cdot)$ for P_Y -almost every y .*

Remark 4.5 (Why the topological hypothesis is unavoidable). Without a hypothesis like “standard Borel” the r.c.d. can fail to exist: one can patch together the a.s.-defined versions $\mathbb{P}(X \in A | Y)$ over a countable generating algebra, but extending to a measure for *every* y requires controlling uncountably many null sets at once, which needs the regularity that Polish spaces provide (via, e.g., the existence of a countable basis and the Carathéodory extension along a compact class). For the ML reader the practical upshot is reassuring: every space that occurs in practice — $\mathbb{R}^d$, compact manifolds, sequence spaces, $\{0, 1\}^n$, spaces of measures with the weak topology — is standard Borel, so the conditional distributions written in papers do exist. The hypothesis is exactly the “standing convention” announced in the Preface.

Theorem 4.6 (Disintegration). *Let μ be the joint law of (X, Y) on $\mathcal{X} \times \mathcal{Y}$ with $\mathcal{X}$ standard Borel, and let $\nu = P_Y$ be the Y -marginal. Then there is a ν -a.e.-unique Markov kernel $\kappa(y, dx)$ with*

$$\mu(dx, dy) = \kappa(y, dx) \nu(dy), \quad (4.4)$$

meaning $\int f d\mu = \int_{\mathcal{Y}} \int_{\mathcal{X}} f(x, y) \kappa(y, dx) \nu(dy)$ for all bounded measurable f . The kernel κ is the r.c.d. of X given Y .

Disintegration (4.4) is the theorem that justifies writing a joint density as “conditional $\times$ marginal” — the single most common factorization in machine learning — at the level of measures, before any density exists. With it, conditional expectation is recovered as an honest integral against a measure:

$$\mathbb{E}[X | Y = y] = \int_{\mathcal{X}} x \kappa(y, dx), \quad \mathbb{E}[\varphi(X) | Y = y] = \int_{\mathcal{X}} \varphi(x) \kappa(y, dx), \quad (4.5)$$

valid for P_Y -a.e. y whenever the integrals converge. This is the form in which a mathematician should *read* every “ $\mathbb{E}_{x \sim p(\cdot | y)}[\varphi(x)]$ ” in a paper.

Chapter 5

Conditional densities and Bayes' rule

A density is a derivative of one measure with respect to another. ML uses the letter p for all of them, so the reference measure must be supplied by the reader.

5.1 When a conditional density exists

Definition 5.1 (Conditional density). Fix a σ -finite reference measure λ on $\mathcal{X}$ (Lebesgue on $\mathbb{R}^d$, counting on a discrete set). Suppose the joint law of (X, Y) has a density $p_{X,Y}$ with respect to $\lambda \otimes \rho$ for some σ -finite ρ on $\mathcal{Y}$, and set $p_Y(y) = \int_{\mathcal{X}} p_{X,Y}(x, y) \lambda(dx)$, the Y -marginal density w.r.t. ρ . Then for y with $p_Y(y) > 0$ define

$$p_{X|Y}(x|y) := \frac{p_{X,Y}(x, y)}{p_Y(y)}, \quad (5.1)$$

and (arbitrarily, e.g. as a fixed reference density) where $p_Y(y) = 0$.

Proposition 5.2 (The density is a density of the r.c.d.). *Under Definition 5.1, the kernel $\kappa(y, A) = \int_A p_{X|Y}(x|y) \lambda(dx)$ is a regular conditional distribution of X given Y . In particular $\{y : p_Y(y) = 0\}$ is P_Y -null, so the arbitrary choice there is immaterial, and (4.5) becomes $\mathbb{E}[\varphi(X) | Y = y] = \int \varphi(x) p_{X|Y}(x|y) \lambda(dx)$.*

Proof. Verify (4.3): for $A \in \mathcal{B}_X$, $B \in \mathcal{B}_Y$, Tonelli gives $\int_B \int_A p_{X|Y}(x|y) \lambda(dx) p_Y(y) \rho(dy) = \int_{A \times B} p_{X,Y} d(\lambda \otimes \rho) = \mathbb{P}(X \in A, Y \in B)$, using $p_{X|Y} p_Y = p_{X,Y}$ off the null set $\{p_Y = 0\}$. $\square$

Warning 5.3 (The same letter p is three different functions). In (5.1) the symbols $p_{X,Y}$, p_Y , and $p_{X|Y}$ are distinct functions on distinct domains; papers drop the subscripts and write $p(x, y)$, $p(y)$, $p(x|y)$, relying on the *argument names* to disambiguate. Thus in ML “ $p(y)$ ” and “ $p(x)$ ” may denote different functions of a real variable. A mathematician should mentally restore the subscripts: the bar in $p(x|y)$ encodes “the $\mathcal{X}$ -density of the r.c.d. at parameter y ,” not a quotient of values of one fixed function.

5.2 Bayes' rule as an identity of densities

Proposition 5.4 (Bayes' rule). *With the hypotheses of Definition 5.1 and $p_X(x) > 0$, where $p_X(x) = \int p_{X,Y}(x, y) \rho(dy)$ is the X -marginal,*

$$p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y) p_Y(y)}{p_X(x)} = \frac{p_{X|Y}(x|y) p_Y(y)}{\int_{\mathcal{Y}} p_{X|Y}(x|y') p_Y(y') \rho(dy')}. \quad (5.2)$$

Read with $Y = \theta$ a parameter and $X = \mathcal{D}$ the data, (5.2) is the posterior = likelihood $\times$ prior / evidence formula of Bayesian ML (Section 8.2). Read with Y a class label it is the generative classifier. The denominator — the *evidence* or *partition function* — is the normalizing integral whose intractability is the reason variational and Monte-Carlo machinery exists at all.

5.3 The Borel–Kolmogorov paradox

We can now state precisely the ambiguity warned of in Chapter 1, and its resolution.

Example 5.5 (Conditioning on a great circle vs. a meridian). Let X be uniform on the unit sphere S^2 . Consider the event “ X lies on a fixed great circle C .” One may present C as {longitude = const} (a meridian) or, after rotating, as {latitude = 0} (the equator). Conditioning X on the meridian, the r.c.d. along C is *not* uniform in arc length (it is weighted by $\cos(\text{latitude})$); conditioning on the equator, it *is* uniform. Both are correct r.c.d.s for their respective *variables*, yet they assign different “conditional laws” to the same null set C .

Remark 5.6 (Diagnosis and resolution). There is no contradiction with Theorem 4.4. An r.c.d. is a kernel attached to a *random variable* Y (longitude, or latitude), not to a bare null event. Uniqueness in Theorem 4.4 is P_Y -a.e. *for that* Y ; changing Y changes the kernel and there is no reason two different conditioning variables should agree on a single fibre that both happen to contain. The lesson for reading ML papers:

“ $p(x|y)$ at a particular y ” is meaningful only relative to the *family* $\{p(\cdot|y)\}_y$ — equivalently, relative to the disintegrating variable Y . Conditioning on an isolated measure-zero event, detached from such a family, is undefined.

This is why diffusion and flow papers are careful to define a whole path of conditional laws $p_t(\cdot|y)$ indexed by t and y , rather than conditioning on individual zero-probability level sets ad hoc.

Part IV

Conditional Independence

Chapter 6

Conditional independence

The symbol $X \perp\!\!\!\perp Y \mid Z$ — “ X and Y are independent given Z ” — is the load-bearing assumption of graphical models, causal inference, and almost every probabilistic factorization in ML. It has several equivalent definitions; knowing which one a paper is using removes most confusion.

6.1 Equivalent definitions

Definition 6.1 (Conditional independence). Let X, Y, Z be random variables into standard Borel spaces. We say X and Y are *conditionally independent given Z* , written $X \perp\!\!\!\perp Y \mid Z$, if any one of the following equivalent conditions holds.

(CI1) (Kernel factorization) a regular conditional distribution of (X, Y) given Z factors:

$$P_{(X,Y) \mid Z}(dx, dy \mid z) = P_{X \mid Z}(dx \mid z) P_{Y \mid Z}(dy \mid z) \quad \text{for } P_Z\text{-a.e. } z. \quad (6.1)$$

(CI2) (Conditional-probability product) for all $A \in \mathcal{B}_X$, $B \in \mathcal{B}_Y$,

$$\mathbb{P}(X \in A, Y \in B \mid Z) \stackrel{\text{a.s.}}{=} \mathbb{P}(X \in A \mid Z) \mathbb{P}(Y \in B \mid Z). \quad (6.2)$$

(CI3) (Conditional-expectation product) for all bounded Borel f, g ,

$$\mathbb{E}[f(X)g(Y) \mid Z] \stackrel{\text{a.s.}}{=} \mathbb{E}[f(X) \mid Z] \mathbb{E}[g(Y) \mid Z]. \quad (6.3)$$

(CI4) (Sufficiency form) for all bounded Borel f ,

$$\mathbb{E}[f(X) \mid Y, Z] \stackrel{\text{a.s.}}{=} \mathbb{E}[f(X) \mid Z]; \quad (6.4)$$

i.e. once Z is known, Y carries no further information about X .

(CI5) (Density factorization, when densities exist)

$$p_{X,Y \mid Z}(x, y \mid z) = p_{X \mid Z}(x \mid z) p_{Y \mid Z}(y \mid z) \quad (\text{a.e.}). \quad (6.5)$$

Proposition 6.2 (Equivalence). *Conditions (CI1)–(CI5) are equivalent (with (CI5) requiring the existence of the relevant densities).*

Notes. (CI1) $\Leftrightarrow$ (CI2) is (4.2) read for the pair (X, Y) and the product kernel. (CI2) $\Leftrightarrow$ (CI3) extends indicators to bounded functions by linearity and monotone convergence. (CI3) $\Rightarrow$ (CI4): for bounded f and any bounded $g(Y), h(Z)$, $\mathbb{E}[\mathbb{E}[f(X) \mid Y, Z] g(Y) h(Z)] = \mathbb{E}[f(X) g(Y) h(Z)] = \mathbb{E}[h(Z) \mathbb{E}[f(X) g(Y) \mid Z]] = \mathbb{E}[h(Z) \mathbb{E}[f(X) \mid Z] \mathbb{E}[g(Y) \mid Z]] = \mathbb{E}[\mathbb{E}[f(X) \mid Z] g(Y) h(Z)]$, using (CI3) and taking out $\mathbb{E}[g(Y) \mid Z]$; since products $g(Y) h(Z)$ generate $\sigma(Y, Z)$ this gives (6.4). (CI4) $\Rightarrow$ (CI3) reverses the computation via the tower rule. (CI5) is (6.1) written through densities (Proposition 5.2). $\square$

Remark 6.3 (The symmetry that the bar hides). Form (6.4) looks asymmetric in X and Y , but conditional independence is symmetric: $X \perp\!\!\!\perp Y \mid Z \iff Y \perp\!\!\!\perp X \mid Z$. The sufficiency reading is nonetheless the one most used in ML — “ Z is a sufficient statistic,” “the representation Z makes the label Y conditionally independent of the input X ,” the Markov property of latent states — so it is worth recognizing it as merely one face of Definition 6.1.

6.2 The graphoid axioms

Conditional independence behaves like a calculus of “irrelevance.” The following identities, the *graphoid axioms*, are the rules that graphical-model algorithms (d-separation, message passing) silently rely on. Write $X \perp\!\!\!\perp Y \mid Z$ for the ternary relation.

Proposition 6.4 (Semi-graphoid axioms). *For random variables W, X, Y, Z :*

$$\text{(Symmetry)} \quad X \perp\!\!\!\perp Y \mid Z \implies Y \perp\!\!\!\perp X \mid Z; \quad (6.6)$$

$$\text{(Decomposition)} \quad X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid Z; \quad (6.7)$$

$$\text{(Weak union)} \quad X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid (Z, W); \quad (6.8)$$

$$\text{(Contraction)} \quad (X \perp\!\!\!\perp Y \mid Z) \wedge (X \perp\!\!\!\perp W \mid (Y, Z)) \implies X \perp\!\!\!\perp (Y, W) \mid Z. \quad (6.9)$$

Proposition 6.5 (Intersection, under positivity). *If the joint density is strictly positive (more generally, under a suitable positivity condition), the additional intersection axiom holds:*

$$(X \perp\!\!\!\perp Y \mid (Z, W)) \wedge (X \perp\!\!\!\perp W \mid (Z, Y)) \implies X \perp\!\!\!\perp (Y, W) \mid Z. \quad (6.10)$$

Warning 6.6 (Intersection can fail). The intersection axiom is the one that requires a hypothesis. Without positivity it is false: take $Y = W$ a single fair coin and $X = Y$; then $X \perp\!\!\!\perp Y \mid (Z, W)$ and $X \perp\!\!\!\perp W \mid (Z, Y)$ hold trivially (the conditioning already determines X), yet X is not independent of (Y, W) given Z . ML papers that invoke “by the graphoid axioms” while manipulating sparse or deterministic models are implicitly assuming the positivity that licenses intersection; a careful reader checks it.

6.3 Factorization over graphs

Definition 6.7 (Bayesian-network factorization). Let G be a directed acyclic graph on vertices $\{1, \dots, d\}$ with parent sets $\text{pa}(i)$. A law p on $\mathcal{X}_1 \times \dots \times \mathcal{X}_d$ *factorizes according to G* if

$$p(x_1, \dots, x_d) = \prod_{i=1}^d p(x_i \mid x_{\text{pa}(i)}). \quad (6.11)$$

Theorem 6.8 (Factorization $\Leftrightarrow$ Markov property). *A positive law factorizes as (6.11) if and only if it satisfies the local Markov property of G : each variable is conditionally independent of its non-descendants given its parents, $X_i \perp\!\!\!\perp X_{\text{nd}(i) \setminus \text{pa}(i)} \mid X_{\text{pa}(i)}$. Either condition is equivalent to the global Markov property (d -separation $\Rightarrow$ conditional independence).*

Equation (6.11) is the rigorous meaning of the “plate diagrams” and factor graphs that open most probabilistic-ML papers, and each factor $p(x_i \mid x_{\text{pa}(i)})$ is a Markov kernel in the sense of Definition 4.1. The autoregressive models of Section 9.1 are the special case of the complete DAG with $\text{pa}(i) = \{1, \dots, i-1\}$, where (6.11) is not an assumption at all but the chain rule of probability.

Part V

Conditional Information Measures

Chapter 7

Relative entropy and the conditional information measures

Every information-theoretic quantity in ML is built from one primitive — relative entropy — so we start there and obtain entropy, cross-entropy, mutual information, and their conditional versions as corollaries. Logarithms are natural unless a paper says “bits.”

7.1 Relative entropy (KL divergence)

Definition 7.1 (Kullback–Leibler divergence). For probability measures P, Q on a common $(\mathcal{X}, \mathcal{B})$,

$$\text{KL}(P \| Q) := \begin{cases} \int_{\mathcal{X}} \log \frac{dP}{dQ} dP = \int_{\mathcal{X}} \frac{dP}{dQ} \log \frac{dP}{dQ} dQ, & P \ll Q, \\ +\infty, & P \not\ll Q. \end{cases} \quad (7.1)$$

If P, Q have densities p, q w.r.t. a common reference λ , then $\text{KL}(P \| Q) = \int p \log(p/q) d\lambda$.

Proposition 7.2 (Gibbs’ inequality). $\text{KL}(P \| Q) \geq 0$, with equality iff $P = Q$. Consequently KL is a divergence: it separates points. It is not a metric — it is asymmetric and violates the triangle inequality.

Proof. Apply Jensen to the strictly convex $-\log$:

$$-\text{KL}(P \| Q) = \int \log \frac{dQ}{dP} dP \leq \log \int \frac{dQ}{dP} dP = \log Q(\mathcal{X}) = 0,$$

with equality iff dQ/dP is P -a.s. constant, i.e. $P = Q$. □

Warning 7.3 (The absolute-continuity trap). $\text{KL}(P \| Q) = +\infty$ whenever P places mass where Q does not. This is the rigorous reason a generative model with a too-small support, or a variational posterior q that is more spread out than the true posterior, incurs the notorious “mode covering vs. mode seeking” asymmetry: $\text{KL}(q \| p)$ (used in variational inference, Section 8.3) blows up if q has mass where $p \approx 0$, whereas $\text{KL}(p \| q)$ (used in maximum likelihood, Section 8.1) blows up if q misses mass that p has. Same symbol KL, opposite failure modes, decided entirely by argument order.

7.2 Entropy and cross-entropy

Definition 7.4 (Entropy and cross-entropy). For a discrete law P with pmf p on a countable $\mathcal{X}$,

$$H(P) = H(X) := -\sum_x p(x) \log p(x) = -\mathbb{E}_P[\log p(X)], \quad (7.2)$$

and for $P \ll \lambda$ with density p (e.g. $\lambda = \text{Lebesgue}$), the *differential entropy*

$$h(X) := - \int_{\mathcal{X}} p(x) \log p(x) \lambda(\mathrm{d}x). \quad (7.3)$$

The *cross-entropy* of Q (density q) relative to P is $H(P, Q) := -\mathbb{E}_P[\log q(X)]$.

Proposition 7.5 (The bookkeeping identity). *Whenever the terms are finite,*

$$H(P, Q) = H(P) + \text{KL}(P \parallel Q). \quad (7.4)$$

Equation (7.4) is why “minimizing cross-entropy” and “minimizing KL to the data distribution” are the same training objective: $H(P)$ does not depend on the model Q , so $\arg \min_Q H(P, Q) = \arg \min_Q \text{KL}(P \parallel Q)$. We return to this in Section 8.1.

Warning 7.6 (Differential entropy is not entropy). For continuous X , $h(X)$ can be negative, is *not* invariant under smooth reparameterization (under $X \mapsto T(X)$ it changes by $\mathbb{E}[\log |\det \nabla T|]$), and is not the limit of discrete entropies of finer quantizations (that limit diverges). It is best regarded only as a term that makes other quantities — mutual information, KL — behave well, since those are differences in which the offending Jacobians and infinities cancel. A mathematician should distrust any standalone use of h .

7.3 Conditional entropy

Definition 7.7 (Conditional entropy). For discrete (X, Y) ,

$$H(X \mid Y) := \sum_y p_Y(y) H(X \mid Y = y) = - \sum_{x,y} p_{X,Y}(x, y) \log p_{X \mid Y}(x \mid y), \quad (7.5)$$

where $H(X \mid Y = y)$ is the entropy of the fibre law $P_{X \mid Y}(\cdot \mid y)$. Equivalently,

$$H(X \mid Y) = \mathbb{E}_Y[H(P_{X \mid Y}(\cdot \mid Y))].$$

So conditional entropy is the *expected entropy of the regular conditional distribution* — an average over Y of an entropy computed inside each fibre, exactly the disintegration (4.4) at work. It is *not* the entropy of a conditional law at a single y ; the single- y object is $H(X \mid Y = y)$, and the unadorned $H(X \mid Y)$ has already integrated it out.

Proposition 7.8 (Chain rule and conditioning reduces entropy). *For discrete variables,*

$$H(X, Y) = H(Y) + H(X \mid Y), \quad 0 \leq H(X \mid Y) \leq H(X), \quad (7.6)$$

the right inequality (“conditioning cannot increase entropy,” with equality iff $X \perp\!\!\!\perp Y$) holding on average only — it can happen that $H(X \mid Y = y) > H(X)$ for particular y .

7.4 Mutual information and its conditional form

Definition 7.9 (Mutual and conditional mutual information).

$$I(X; Y) := \text{KL}(P_{X,Y} \parallel P_X \otimes P_Y), \quad (7.7)$$

$$I(X; Y \mid Z) := \mathbb{E}_Z[\text{KL}(P_{X,Y \mid Z}(\cdot \mid Z) \parallel P_{X \mid Z}(\cdot \mid Z) \otimes P_{Y \mid Z}(\cdot \mid Z))]. \quad (7.8)$$

Definition (7.7) is reference-measure-free and so escapes the pathologies of differential entropy (Warning 7.6): mutual information is a KL divergence between honest measures and is therefore nonnegative, invariant under bijective reparameterization of X and of Y , and equal to $+\infty$ exactly when $P_{X,Y} \not\ll P_X \otimes P_Y$. The conditional version (7.8) is, once more, an *average over Z* of a per-fibre KL.

Proposition 7.10 (Identities). *When the entropies are finite,*

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) \geq 0, \quad (7.9)$$

$$I(X; Y | Z) = H(X | Z) - H(X | Y, Z) \geq 0, \quad (7.10)$$

$$I(X; Y, Z) = I(X; Z) + I(X; Y | Z) \quad (\text{chain rule}). \quad (7.11)$$

Moreover $I(X; Y | Z) = 0 \iff X \perp\!\!\!\perp Y | Z$, tying Part IV to Part V: conditional independence is exactly the vanishing of conditional mutual information.

Remark 7.11 (Where these appear). Conditional mutual information $I(X; Y | Z)$ is the quantity behind the information-bottleneck objective, the “minimality/sufficiency” analyses of learned representations, and the conditional-independence tests of causal discovery. The chain rule (7.11) is the information-theoretic shadow of the probability chain rule that drives autoregressive models (Section 9.1). Recognizing all of these as averaged per-fibre KL divergences — never as arithmetic on single- y entropies — is the main thing a mathematician needs to read them safely.

7.5 Conditional KL and the chain rule for divergence

Definition 7.12 (Conditional / expected KL). Given kernels $P_{X|Y}, Q_{X|Y}$ and a law P_Y ,

$$\text{KL}(P_{X|Y} \| Q_{X|Y} | P_Y) := \int_Y \text{KL}(P_{X|Y}(\cdot | y) \| Q_{X|Y}(\cdot | y)) P_Y(dy). \quad (7.12)$$

Proposition 7.13 (Chain rule for KL). *For joint laws $P = P_Y P_{X|Y}$ and $Q = Q_Y Q_{X|Y}$,*

$$\text{KL}(P_{X,Y} \| Q_{X,Y}) = \text{KL}(P_Y \| Q_Y) + \text{KL}(P_{X|Y} \| Q_{X|Y} | P_Y). \quad (7.13)$$

The conditional KL (7.12) — note it averages the second argument’s null-set behavior over P_Y , the *first* law’s marginal — is the per-step training signal of diffusion models, where the loss is a sum of terms $\text{KL}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$ averaged over the forward process (Section 9.2). The chain rule (7.13) is what telescopes those per-step divergences into a bound on the divergence between full trajectory laws.

Part VI

The Machine-Learning Symbol Dictionary

This part is the reverse map. Each entry takes a symbol *as it is actually printed in a thesis*, states what a reader should mentally substitute, and points to the rigorous object in Parts I–V. The recurring move is always one of three: a bar denotes a Markov kernel (Definition 4.1), a conditional expectation (Definition 2.1), or a conditional density (Definition 5.1); everything else is bookkeeping.

Chapter 8

Likelihood, posterior, and the variational bar

8.1 Likelihood, NLL, and MLE as a KL projection

Symbol. $p_\theta(x)$, $p_\theta(y|x)$, $\mathcal{L}(\theta) = \prod_i p_\theta(x_i)$, $-\log p_\theta(y|x)$.

A *model* is a parameterized family of probability measures $\{P_\theta : \theta \in \Theta\}$ on the data space, each with density p_θ w.r.t. a fixed reference λ . Read $p_\theta(x)$ as “the density of P_θ at x ,” a function of x for fixed θ ; read the *likelihood* $\theta \mapsto p_\theta(x)$ as the same expression viewed as a function of θ for fixed observed x . The *conditional likelihood* $p_\theta(y|x)$ is a Markov kernel (Definition 4.1) indexed by θ : a discriminative model predicting y from x . The *negative log-likelihood* (NLL) $-\log p_\theta(y|x)$ is the per-example loss of supervised learning.

Proposition 8.1 (Maximum likelihood is KL projection onto the model). *Let $\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ be the empirical distribution of i.i.d. data $x_i \sim P_\star$. Then*

$$\arg \max_{\theta} \frac{1}{n} \sum_{i=1}^n \log p_\theta(x_i) = \arg \min_{\theta} H(\hat{P}_n, P_\theta) = \arg \min_{\theta} \text{KL}(\hat{P}_n \| P_\theta), \quad (8.1)$$

and as $n \rightarrow \infty$ the population objective is $\arg \min_{\theta} \text{KL}(P_\star \| P_\theta)$.

Proof. Maximizing the average log-likelihood equals minimizing the cross-entropy $H(\hat{P}_n, P_\theta) = -\frac{1}{n} \sum_i \log p_\theta(x_i)$; subtract the θ -free term $H(\hat{P}_n)$ and apply (7.4). $\square$

So “train by minimizing cross-entropy” is “project the data distribution onto the model in KL” — the forward $\text{KL}(P_\star \| P_\theta)$, the mode-covering one of Warning 7.6’s companion. The bar in $p_\theta(y|x)$ contributes the conditional version: minimizing $\mathbb{E}_x[\text{KL}(P_\star(\cdot|x) \| P_\theta(\cdot|x))]$, the conditional KL (7.12) of Definition 7.12.

8.2 Posterior and evidence

Symbol. $p(\theta|\mathcal{D})$, $p(\mathcal{D}|\theta)$, $p(\mathcal{D}) = \int p(\mathcal{D}|\theta)p(\theta) d\theta$.

In the Bayesian picture the parameter θ is itself a random variable with prior law $p(\theta)$. The *posterior* $p(\theta|\mathcal{D})$ is literally the conditional density of Definition 5.1 with $X = \theta$, $Y = \mathcal{D}$, supplied by Bayes’ rule (Proposition 5.4):

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}, \quad p(\mathcal{D}) = \int_{\Theta} p(\mathcal{D}|\theta)p(\theta) d\theta. \quad (8.2)$$

The normalizer $p(\mathcal{D})$ — the *evidence* or *marginal likelihood* — is the $\mathcal{D}$ -marginal density and is the integral whose intractability motivates everything in the next section.

Warning 8.2 ($p(x; \theta)$ vs. $p(x | \theta)$: semicolon is not a bar). A frequentist writes $p(x; \theta)$ or $p_\theta(x)$ — θ is a fixed index, not an event, and there is no measure on Θ . A Bayesian writes $p(x | \theta)$ — θ is a random variable and the bar is genuine conditioning. The two coincide *numerically* but mean different things: only the second makes $p(\theta | x)$ meaningful. When a paper slides between $p(x; \theta)$ and $p(x | \theta)$ it is silently choosing whether Θ carries a probability measure. A mathematician should track which.

8.3 The variational posterior and the ELBO

Symbol. $q_\phi(z | x)$, $\mathbb{E}_{q_\phi(z | x)}[\cdot]$, ELBO.

A latent-variable model posits $p_\theta(x, z) = p_\theta(x | z)p(z)$ with a prior $p(z)$ and a decoder kernel $p_\theta(x | z)$. The exact posterior $p_\theta(z | x) = p_\theta(x, z)/p_\theta(x)$ is intractable because $p_\theta(x) = \int p_\theta(x | z)p(z) dz$ is. The *variational posterior* (or encoder, or recognition kernel) $q_\phi(z | x)$ is a second Markov kernel, parameterized by ϕ , introduced to approximate it. The expectation $\mathbb{E}_{q_\phi(z | x)}[\cdot]$ is the integral $\int(\cdot) q_\phi(dz | x)$ against that kernel — a conditional expectation under q , in the sense of (4.5).

Proposition 8.3 (Evidence lower bound and its gap). *For any kernel $q_\phi(\cdot | x) \ll p_\theta(\cdot | x)$,*

$$\log p_\theta(x) = \underbrace{\mathbb{E}_{q_\phi(z | x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z | x)} \right]}_{\text{ELBO}(\theta, \phi; x)} + \text{KL}(q_\phi(\cdot | x) \| p_\theta(\cdot | x)), \quad (8.3)$$

and since the KL term is ≥ 0 , $\text{ELBO}(\theta, \phi; x) \leq \log p_\theta(x)$ with equality iff $q_\phi(\cdot | x) = p_\theta(\cdot | x)$. Equivalently

$$\text{ELBO}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z | x)}[\log p_\theta(x | z)] - \text{KL}(q_\phi(\cdot | x) \| p(\cdot)). \quad (8.4)$$

Proof. Insert $1 = q_\phi(z | x)/q_\phi(z | x)$ into $\log p_\theta(x) = \log \int p_\theta(x, z) dz$ is not valid directly; instead write $\log p_\theta(x) = \mathbb{E}_{q_\phi(z | x)}[\log p_\theta(x)]$ (the integrand is constant in z), expand $\log p_\theta(x) = \log p_\theta(x, z) - \log p_\theta(z | x)$, and add and subtract $\log q_\phi(z | x)$. The two resulting expectations are the ELBO and the KL. Equation (8.4) regroups $p_\theta(x, z) = p_\theta(x | z)p(z)$. $\square$

Remark 8.4 (Three bars, three different objects, one equation). Equation (8.3) contains the encoder kernel $q_\phi(z | x)$, the decoder kernel $p_\theta(x | z)$, and the exact-posterior kernel $p_\theta(z | x)$ — three distinct Markov kernels, written with the same bar. The variational gap is the *reverse* KL $\text{KL}(q_\phi \| p_\theta(\cdot | x))$, the mode-seeking direction; this is why VAEs are prone to posterior collapse and under-dispersed encoders. The mathematician’s reading of “maximize the ELBO” is: minimize, over ϕ , the reverse conditional KL between two kernels, simultaneously fitting θ by a lower bound on the marginal log-likelihood.

Chapter 9

Sequential, diffusion, and reinforcement bars

9.1 Autoregressive models: the chain rule is not an assumption

Symbol. $p(x_{1:T}) = \prod_{t=1}^T p(x_t | x_{<t})$.

For any random vector $(X_1, \dots, X_T)$ into standard Borel spaces, the *chain rule of probability* gives regular conditional distributions $P_{X_t | X_{<t}}$ with

$$P_{X_{1:T}}(\mathrm{d}x_{1:T}) = \prod_{t=1}^T P_{X_t | X_{<t}}(\mathrm{d}x_t | x_{<t}), \quad (9.1)$$

by iterating disintegration (Theorem 4.6) $T - 1$ times. Unlike a graphical-model factorization, (9.1) holds for *every* joint law: it is an identity, not a modeling assumption. An “autoregressive model” replaces each true kernel by a learned kernel $p_\theta(x_t | x_{<t})$ (the softmax over a vocabulary in a language model, a mixture or a logistic in a density model), and its NLL training loss is the conditional cross-entropy summed over positions.

Remark 9.1. What the model assumes is never the factorization (9.1) but the *parametric form and conditioning set* of each kernel: a finite context window, a particular network, weight sharing across t . The bar in $p(x_t | x_{<t})$ is doing the honest work of Theorem 4.6; the modeling lives in the function class, not in the conditioning.

9.2 Markov chains, transition kernels, and diffusion

Symbol. $q(x_t | x_{t-1}), \quad p_\theta(x_{t-1} | x_t), \quad \nabla_x \log p_t(x), \quad \nabla_x \log p_t(x | y)$.

A (time-inhomogeneous) Markov chain is a prior p_0 together with transition kernels $\kappa_t(x_{t-1}, \mathrm{d}x_t)$; its law is the kernel product of Remark 4.2, and the Markov property is the conditional independence $X_t \perp\!\!\!\perp X_{<t-1} | X_{t-1}$ of Definition 6.1. A *diffusion model* fixes a forward chain $q(x_t | x_{t-1})$ that gradually adds noise and learns a reverse chain $p_\theta(x_{t-1} | x_t)$; the training loss is a sum of conditional KL terms (7.12) $\mathrm{KL}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$ averaged over the forward law, telescoped via the chain rule for KL (7.13) into a bound on $-\log p_\theta(x_0)$.

Definition 9.2 (Score). The (*Stein*) *score* of a density p_t on $\mathbb{R}^d$ is the gradient of its log-density in the *state*, $s_t(x) := \nabla_x \log p_t(x)$ — *not* a derivative in any parameter. It is invariant to the normalizing constant of p_t , which is why score-based models can sidestep intractable partition functions.

Proposition 9.3 (Conditional score and Bayes; the two guidances). *For a joint $p_t(x, y) = p_t(x)p_t(y|x)$ on $\mathbb{R}^d \times \mathcal{Y}$ with y held fixed, differentiating $\log p_t(x|y) = \log p_t(x) + \log p_t(y|x) - \log p_t(y)$ in x gives the conditional score decomposition*

$$\nabla_x \log p_t(x|y) = \underbrace{\nabla_x \log p_t(x)}_{\text{unconditional score}} + \underbrace{\nabla_x \log p_t(y|x)}_{\text{classifier gradient}}, \quad (9.2)$$

since $\nabla_x \log p_t(y) = 0$. Classifier guidance implements (9.2) literally, adding the gradient of an auxiliary classifier $p_t(y|x)$. Classifier-free guidance instead trains one network to output both the conditional and unconditional scores $s_\theta(x, y), s_\theta(x, \emptyset)$ and forms the guided field

$$\tilde{s}_\theta(x, y) = (1 + w) s_\theta(x, y) - w s_\theta(x, \emptyset) = s_\theta(x, y) + w(s_\theta(x, y) - s_\theta(x, \emptyset)), \quad (9.3)$$

with guidance weight $w \geq 0$; the bracket is exactly the classifier gradient $\nabla_x \log p_t(y|x)$ of (9.2), so $w = 0$ recovers the true conditional score and $w > 0$ samples from a sharpened, tempered conditional $\propto p_t(x)p_t(y|x)^{1+w}$.

Remark 9.4 (A bar that is a gradient of a log of a kernel). The symbol $\nabla_x \log p_t(x|y)$ stacks three of this handbook’s constructions: a conditional density $p_t(x|y)$ (Definition 5.1) indexed by a diffusion time t , its logarithm, and a gradient in the *conditioned* variable x . Equation (9.2) is just Bayes’ rule (Proposition 5.4) after a logarithm and a gradient — the single most important “conditional thing” in modern generative modeling, and entirely elementary once each piece is named.

9.3 Policies, value functions, and the Bellman bar

Symbol. $\pi(a|s), \quad V^\pi(s) = \mathbb{E}_\pi[\sum_t \gamma^t r_t | S_0 = s], \quad Q^\pi(s, a).$

A *policy* $\pi(a|s)$ is a Markov kernel from states to actions (Definition 4.1); together with the environment’s transition kernel $P(s'|s, a)$ and reward it defines a law on trajectories. The *value function*

$$V^\pi(s) := \mathbb{E}_\pi\left[\sum_{t \geq 0} \gamma^t R_t \mid S_0 = s\right], \quad Q^\pi(s, a) := \mathbb{E}_\pi\left[\sum_{t \geq 0} \gamma^t R_t \mid S_0 = s, A_0 = a\right], \quad (9.4)$$

is an evaluated conditional expectation (Definition 3.3) of a discounted total reward; “ $| S_0 = s$ ” is the conditioning-on-a- random-variable of Chapter 3, well defined for P_{S_0} -a.e. s .

Proposition 9.5 (Bellman equation as the tower rule). *With the per-step conditional expectation over $A_0 \sim \pi(\cdot|s), S_1 \sim P(\cdot|s, A_0)$,*

$$V^\pi(s) = \mathbb{E}_\pi[R_0 + \gamma V^\pi(S_1) | S_0 = s]. \quad (9.5)$$

Proof. Split the geometric sum in (9.4) as $R_0 + \gamma \sum_{t \geq 0} \gamma^t R_{t+1}$, and apply the tower rule (2.3) conditioning the tail on $(S_1, A_1, \dots)$ through S_1 ; the Markov property collapses the tail’s conditional expectation to $V^\pi(S_1)$. $\square$

So the Bellman equation — the fixed-point identity behind value iteration, Q-learning, and temporal-difference methods — is nothing but the tower property of conditional expectation (Proposition 2.5(4)) applied to a Markov reward process. Every “bootstrapping” target $r + \gamma V(s')$ is a one-step conditional expectation standing in for the full return.

9.4 Filtrations and the martingale bar

Symbol. $\mathcal{F}_t, \mathbb{E}[X_{t+1} | \mathcal{F}_t], \mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t$.

A *filtration* $(\mathcal{F}_t)_{t \geq 0}$ is an increasing family of sub- σ -algebras, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for $s \leq t$: “the information available by time t .” This is conditioning on a σ -algebra (Definition 2.1) in its purest form, with no random variable to disintegrate against. A process is a *martingale* if it is adapted, integrable, and $\mathbb{E}[X_{t+1} | \mathcal{F}_t] \stackrel{\text{a.s.}}{=} X_t$. Stochastic-approximation and SGD convergence proofs are written in this language: the noise sequence is a martingale-difference, $\mathbb{E}[\xi_{t+1} | \mathcal{F}_t] = 0$, and the tower rule plus a supermartingale convergence theorem do the rest.

Remark 9.6 (Why $\mathcal{F}_t$ and not “| the past”). Writing $\mathbb{E}[X_{t+1} | \mathcal{F}_t]$ rather than $\mathbb{E}[X_{t+1} | X_0, \dots, X_t]$ is the deliberately more general move: $\mathcal{F}_t$ may encode side information beyond the past values of X , and the conditional expectation is defined directly on the σ -algebra without ever choosing a conditioning random variable. For a mathematician this is the *cleanest* conditional bar in all of ML — it is exactly Definition 2.1, untranslated.

Chapter 10

A field guide to notational pitfalls

We close the dictionary with the ambiguities that most often trip a mathematician, each cross-referenced to its resolution.

1. **The bar is overloaded.** $\mathbb{P}(A|B)$ conditions on an *event* (Definition 1.1); $\mathbb{E}[X|\mathcal{G}]$ on a σ -algebra (Definition 2.1); $\mathbb{E}[X|Y]$ on a *random variable* (Chapter 3); $\mathbb{E}[X|Y=y]$ is a *function evaluated* (Definition 3.3); $p(x|y)$ a *kernel/density* (Definition 5.1). Identify which before reading on.
2. **Random variable vs. number.** $\mathbb{E}[X|Y]$ is a random variable; $\mathbb{E}[X|Y=y]$ is a (version of a) deterministic function. Identities about the latter hold only for P_Y -a.e. y (Warning 3.4).
3. **“=” usually means “ $\stackrel{\text{a.s.}}{=}$ ”.** Conditional objects are equivalence classes modulo null sets. A clean-looking equation between conditional expectations is an almost-sure statement unless a specific version (e.g. a regular kernel) has been fixed.
4. **Same letter p , different functions.** $p(x)$, $p(y)$, $p(x, y)$, $p(x|y)$ are distinct functions disambiguated only by their argument names. Restore subscripts $p_X, p_Y, p_{X,Y}, p_{X|Y}$ when in doubt.
5. **Semicolon vs. bar.** $p(x; \theta)$ / $p_\theta(x)$ (frequentist index, no measure on Θ) vs. $p(x|\theta)$ (Bayesian, θ random). Only the latter licenses a posterior $p(\theta|x)$.
6. **Reference measure is implicit.** “Density” always means “density with respect to *some* λ ” — usually Lebesgue or counting. Differential entropy and change-of-variables (normalizing flows!) are sensitive to this choice; mutual information and KL are not (Warning 7.6).
7. **Conditioning on a null event needs a family.** A single $p(\cdot|Y=y)$ is meaningless detached from the disintegrating variable Y ; the Borel–Kolmogorov paradox (Section 5.3) is the price of forgetting this.
8. **KL direction matters.** $\text{KL}(p\|q)$ (maximum likelihood, mode-covering) and $\text{KL}(q\|p)$ (variational inference, mode-seeking) are different objectives with different failure modes (Warning 7.6’s box, Remark 8.3’s discussion).
9. **∇ in which variable?** In $\nabla_x \log p_t(x|y)$ the gradient is in the *state* x (a score), not in parameters θ ; in $\nabla_\theta \log p_\theta(x)$ it is in parameters (for gradient ascent). The subscript on ∇ is not decoration.
10. **Expectation subscripts name a measure.** $\mathbb{E}_{x \sim p(\cdot|y)}[\cdot]$ integrates against the kernel $p(\cdot|y)$ at fixed y (a conditional expectation); $\mathbb{E}_{x,y}$ integrates against the joint. Reading off the subscript tells you which measure and hence which Fubini step is legal.

Appendices

Appendix A

The theorems relied upon

For completeness we record, without proof, the three measure-theoretic results that do all the heavy lifting above, plus the topological hypothesis under which conditional distributions exist. Standard references are Durrett, *Probability: Theory and Examples*; Kallenberg, *Foundations of Modern Probability*; and Çinlar, *Probability and Stochastics*.

Theorem A.1 (Radon–Nikodym). *Let μ, ν be σ -finite measures on $(\Omega, \mathcal{F})$ with $\nu \ll \mu$ (i.e. $\mu(A) = 0 \Rightarrow \nu(A) = 0$). Then there is an $\mathcal{F}$ -measurable $f \geq 0$, unique up to μ -a.e. equality, with $\nu(A) = \int_A f \, d\mu$ for all $A \in \mathcal{F}$. One writes $f = d\nu/d\mu$.*

This is the engine of Theorem 2.2: conditional expectation is a Radon–Nikodym derivative, taken on the sub- σ -algebra $\mathcal{G}$. Conditional densities (Definition 5.1) and the KL integrand $\log(dP/dQ)$ (Definition 7.1) are likewise Radon–Nikodym derivatives.

Theorem A.2 (Doob–Dynkin lemma). *Let $Y : \Omega \rightarrow \mathcal{Y}$ and $Z : \Omega \rightarrow \mathbb{R}$ be measurable with $\mathcal{Y}$ standard Borel. Then Z is $\sigma(Y)$ -measurable iff $Z = g \circ Y$ for some Borel $g : \mathcal{Y} \rightarrow \mathbb{R}$.*

This is what turns $\mathbb{E}[X | Y]$ (a $\sigma(Y)$ -measurable random variable) into a function g of y , hence into the symbol $\mathbb{E}[X | Y = y]$ and the regression function (Chapter 3).

Definition A.3 (Standard Borel space). A measurable space $(\mathcal{X}, \mathcal{B})$ is *standard Borel* if it is Borel isomorphic to a Borel subset of a Polish (separable, completely metrizable) space with its Borel σ -algebra. Equivalently, it is isomorphic to one of: a finite or countable discrete set, or $\mathbb{R}$ (the Borel isomorphism theorem). All of $\mathbb{R}^d$, separable Banach and Hilbert spaces, compact metric spaces, $\{0, 1\}^{\mathbb{N}}$, and spaces of probability measures with the weak topology are standard Borel.

Theorem A.4 (Existence of regular conditional distributions / disintegration). *Let $X : \Omega \rightarrow \mathcal{X}$, $Y : \Omega \rightarrow \mathcal{Y}$ be random variables with $\mathcal{X}$ standard Borel. Then a regular conditional distribution $\kappa(y, dx)$ of X given Y exists (Definition 4.3), is P_Y -a.e. unique, and disintegrates the joint law as $P_{X,Y}(dx, dy) = \kappa(y, dx) P_Y(dy)$ (Theorem 4.6). The standard-Borel hypothesis on $\mathcal{X}$ is essential; it fails for general measurable spaces.*

This single hypothesis — announced as a standing convention in the Preface and satisfied by every space occurring in practice — is what makes the entire conditional vocabulary of machine learning well defined. With it, $p(x | y)$, $\pi(a | s)$, $q_\phi(z | x)$, and every other bar in this handbook denotes an object that provably exists.

Appendix B

Notation cheat sheet

Symbol (as printed)	Rigorous meaning / where defined
$\mathbb{P}(A B)$	Elementary conditional probability, $\mathbb{P}(A \cap B)/\mathbb{P}(B)$, needs $\mathbb{P}(B) > 0$. Def. 1.1.
$\mathbb{E}[X \mathcal{G}]$	Conditional expectation: the $\mathcal{G}$ -measurable r.v. with matching local averages; a Radon–Nikodym derivative; the L^2 -projection onto $L^2(\mathcal{G})$. Def. 2.1.
$\mathbb{E}[X Y]$	$= \mathbb{E}[X \sigma(Y)]$, a $\sigma(Y)$ -measurable random variable; equals $g(Y)$ by Doob–Dynkin. Ch. 3.
$\mathbb{E}[X Y = y]$	$= g(y)$, a deterministic function defined for P_Y -a.e. y only. Def. 3.3, Warning 3.4.
$\mathbb{P}(A \mathcal{G}), \mathbb{P}(A Y)$	$\mathbb{E}[\mathbf{1}_A \mathcal{G}]$; per-event a.s.-defined random variable. Def. 2.7.
$\text{Var}(X Y)$	Conditional variance; obeys the law of total variance (2.8). Def. 2.8.
$P_{X Y}(\cdot y), \kappa(y, dx)$	Regular conditional distribution / Markov kernel; exists on standard Borel spaces. Defs. 4.1, 4.3; Thm. 4.4.
$p(x y)$	Conditional density $= p_{X,Y}(x, y)/p_Y(y)$ w.r.t. a reference measure; a density of the r.c.d. Def. 5.1.
$p(\theta \mathcal{D})$	Posterior $= \text{likelihood} \times \text{prior} / \text{evidence}$ (Bayes’ rule). Prop. 5.4; §8.2.
$X \perp\!\!\!\perp Y Z$	Conditional independence; kernel/conditional-expectation/density factorization; $\iff I(X; Y Z) = 0$. Def. 6.1.
$H(X Y)$	Conditional entropy $= \mathbb{E}_Y[H(P_{X Y}(\cdot Y))]$; expected entropy of the r.c.d. Def. 7.7.
$I(X; Y Z)$	Conditional mutual information; expected per-fibre KL; ≥ 0 , $= 0 \iff X \perp\!\!\!\perp Y Z$. Def. 7.9.
$\text{KL}(P \ Q)$	Relative entropy $\int \log(dP/dQ) dP$ if $P \ll Q$, else $+\infty$; asymmetric. Def. 7.1.
$\text{KL}(P_{X Y} \ Q_{X Y} P_Y)$	Conditional / expected KL; the per-step diffusion loss. Def. 7.12.

Symbol (as printed)	Rigorous meaning / where defined
$q_\phi(z x)$	Variational posterior / encoder kernel; approximates $p_\theta(z x)$. §8.3.
$\mathbb{E}_{q_\phi(z x)}[\cdot]$	Integral against the kernel $q_\phi(\cdot x)$ at fixed x ; a conditional expectation under q . Eq. (4.5).
$\prod_t p(x_t x_{<t})$	Chain rule of probability (an identity, by iterated disintegration), not a modeling assumption. §9.1.
$\prod_i p(x_i x_{\text{pa}(i)})$	Bayesian-network factorization over a DAG (a Markov assumption). Def. 6.7; Thm. 6.8.
$q(x_t x_{t-1}), p_\theta(x_{t-1} x_t)$	Forward / reverse diffusion transition kernels (Markov chain). §9.2.
$\nabla_x \log p_t(x)$	Stein score (gradient in state, not parameters); normalizer-free. §9.2.
$\nabla_x \log p_t(x y)$	Conditional score = $\nabla_x \log p_t(x) + \nabla_x \log p_t(y x)$ (Bayes). Prop. 9.3.
$\pi(a s)$	Policy: a Markov kernel from states to actions. §9.3.
$V^\pi(s) = \mathbb{E}_\pi[\sum_t \gamma^t R_t S_0 = s]$	Value: evaluated conditional expectation of discounted return; Bellman = tower rule. Eqs. (9.4), (9.5).
$\mathbb{E}[X_{t+1} \mathcal{F}_t]$	Conditioning on a filtration σ -algebra; martingale if $\stackrel{\text{a.s.}}{=} X_t$. §10 entry, Def. 2.1.

One rule to carry away: a bar is always one of three things — a conditional expectation (a projection / Radon–Nikodym derivative on a σ -algebra), a Markov kernel (a measurably-indexed family of measures), or a conditional density (the Radon–Nikodym derivative of that kernel against a reference measure). Decide which, and the notation stops being ambiguous.